

JUAN SEBASTIAN ZAPATA ACOSTA

Bogotá, Colombia

Microbiologist | MSc Candidate in Computational Biology | Bioinformatics & Data Science

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PROFESSIONAL SUMMARY

Microbiologist and MSc candidate in Computational Biology with experience spanning microbiology, bioinformatics, biological data analysis, process optimization, applied research, metagenomics, and laboratory operations management. Experienced in developing and applying computational workflows for microbial community analysis, functional annotation, metagenome assembly, and predictive biological modeling. Skilled in R, Python, statistical analysis, and scientific data visualization, with demonstrated experience leading laboratory teams, managing quality systems, and translating complex biological datasets into actionable insights for research and industrial applications.

PROFESSIONAL EXPERIENCE

FLORES SAGARO S.A.

Microbiology and ground laboratory Chief | May 2025 – November 2026

- Coordinate technological workflows for high-scale agricultural bioinput development and plant-soil microbiome optimization.
- Lead, trained and supervised an 8-person laboratory team, improving workflow organization and technical consistency.
- Manage microbiological monitoring, technical documentation, quality records, and production data.
- Analyze microbiological and operational information to support decision-making.

Achievements:

- Implemented the laboratory's first environmental microbiological quality control program, including routine monitoring protocols and contamination surveillance.
- Standardized and optimized microbial production protocols, improving process reproducibility and operational efficiency.
- Developed laboratory SOPs, quality records, production forms, and technical documentation supporting standardized operations.

GIAHB (STARTUP)

Technical Development Director | June 2025 - Present

- Develop emerging technologies for future commercialization.
- Lead problem-solving processes applied to the industry from a technical and specialized position within the development area.
- Manage documentation, technological advancements, and product proposals to generate social impact.

Achievements:

- Consolidated a simulated product as the first prototype of the product to be commercialized.
- Developed collaborations with entities such as the University of the Andes to support and develop the brand's final product.

TECMOL PHARMACEUTICALS

Microbiology Analyst | March 2025 – April 2025

- Conducted microbiological quality assessments for pharmaceutical, agronomic, and veterinary products.
- Performed microbiological testing and analysis following USP pharmacopeia standards and standardized laboratory procedures.
- Supported quality assurance activities through documentation review, data recording, and compliance-oriented workflows.

Achievements:

- Successfully performed microbiological quality testing following USP guidelines while maintaining regulatory compliance.
- Supported contamination control programs ensuring microbiological product safety.
- Maintained complete laboratory documentation supporting quality assurance and traceability.

BBC BREWERY DE LA SABANA S.A.S.

Microbiologist – Quality Analyst | November 2024 – February 2025

- Performed microbiological monitoring across multiple production stages; fermentation, malting, packaging and finished product.
- Conducted microbial enumeration and quality control testing to support process consistency and product safety.
- Maintained technical documentation, traceability records, and quality-related reports under standardized operational procedures.

Achievements:

- Supported microbiological quality assurance across product stages.
- Assisted identifying production deviations through microbiological monitoring.
- Contributed to maintaining product consistency under industrial quality standards.

CONCENTRIX

Bilingual Customer Support Agent - Bogotá, Colombia | February 2024 – June 2024

- Delivered customer support in English and Spanish within an international environment.
- Managed documentation, issue tracking, and workflow organization.
- Worked under KPI-based systems and performance metrics.

Achievements:

- Developed strong communication, organization, and remote collaboration skills.
- Managed clients and platforms of credit, billing and remote work.

UNIVERSITY OF LOS ANDES – BIOPHYSICS INVESTIGATION GROUP

Research Assistant | 2021 – 2023

- Processed and analyzed biological datasets using bioinformatics and computational tools.
- Supported experimental data interpretation through data processing and analytical workflows.

- Worked with biological samples, laboratory instrumentation, and interdisciplinary research methodologies.
- Collaborated with research teams integrating laboratory and computational approaches.

Achievements:

- Developed computational workflows that improved biological data processing efficiency.
- Contributed to interdisciplinary projects integrating experimental microbiology and computational biology.
- Generated reproducible research datasets supporting ongoing scientific projects.

EDUCATION

UNIVERSIDAD DE LOS ANDES MSc in Computational Biology (Ongoing) | Bogotá, Colombia

UNIVERSIDAD DE LOS ANDES BSc in Microbiology | Bogotá, Colombia | January 2018 – March 2024

PROJECTS

Development of a Functional Consortium to Generate Compost from Coffee Waste and the Metagenomic Characterization of the Intestinal Microbiota of *Eisenia fetida* | Master's Project Computational Biology at the University of the Andes | January 2026 – Present

Research project focused on the taxonomic and functional characterization of microbial communities associated with the gut of *Eisenia fetida* using shotgun sequencing and bioinformatics tools

- Reconstruction of microbial genomes (MAGs) from shotgun metagenomic sequencing.
- Functional annotation of microbial communities.
- Metabolic pathway reconstruction and ecological interaction analysis.
- Development of reproducible bioinformatics workflows using Python and R.

Volunteer – Palmari Nature Reserve | June 2024- October 2024

- Provided guidance and translation services (Spanish - English or Portuguese - English) for tourists visiting the reserve.
- Characterize and initiate outreach processes regarding the fungal species found in the areas belonging to the reserve.
- I cared for the health of visitors by supporting the proper organization and use of the various medications available at the reserve, filling the role of nurse while I was there.
- I built connections with the inhabitants, guides, and other people associated with the reserve, seeking to foster curiosity and trust in the project.

Gene Expression Analysis in *E. coli* Biofilms | Undergraduate Research Project University of the Andes | January 2021 – December 2023

Research project focused on the computational analysis of biological images to evaluate gene expression patterns associated with cellular stress in *Escherichia coli* communities.

- I processed and analyzed images using the Python programming language.
- I extracted quantitative variables from microscopy data.

- Integrate computational analysis and experimental biology.
- Develop tools for automating biological analyses.

SKILLS

Bioinformatics

- Metagenomics
- Microbiome Analysis
- Functional Annotation
- Computational Biology
- Predictive Biological Modeling

Programming

- Python
- R
- Linux
- Bash

Data Analysis

- Statistical Analysis
- Scientific Data Visualization
- Biological Data Processing

Laboratory

- Laboratory Leadership
- Quality Management Systems
- SOP Development
- Process Optimization
- Microbiological Quality Control

LANGUAGES

- Spanish — Native
- English — B2
- Portuguese — Basic
- German — Basic